

Prince Agrawal

☎ 8265819270 ✉ princeagrawal1013@gmail.com 📁 [Portfolio](#) 🔗 [LinkedIn](#) 🐙 [GitHub](#) </> [LeetCode](#) </> [CodeChef](#)

Education

Indian Institute of Information Technology and Management, Gwalior

Aug 2023 – May 2027

Bachelors of Technology (B.Tech) in Electrical and Electronics Engineering

CGPA: 9.42/10.0

Experience

Software Engineering Intern — JMK Spares and Wears

May 2026 – Present

- Architecting a full-stack **Demand Intelligence Platform** for a 2,900+ client B2B manufacturer, integrating a **Next.js/PostgreSQL** frontend with a **Python ML** backend.
- Designing predictive demand forecasting models utilizing **LightGBM**, **XGBoost**, and **CatBoost**, targeting a **25%** increase in forecast accuracy to anticipate client restocks.
- Building an intelligent **RAG-powered AI Assistant** using **Google Gemini LLM** to process unstructured customer queries, projected to reduce manual resolution time by **40%**.
- Automating the B2B sales pipeline by integrating the **WhatsApp Business API** and **pdf-lib** for automated PDF quote generation, expected to accelerate response times by **50%**.

Projects

CureCart | AI-Powered Healthcare Platform (Next.js, Prisma, Gemini AI)

May 2026 – Present

- Constructed a full-stack healthcare e-commerce platform (**Next.js 15**, **TypeScript**); implemented **SSR** and Edge caching, reducing load times by **40%** and achieving a **95+** Lighthouse score.
- Integrated an intelligent health assistant utilizing **Gemini AI** and a **RAG pipeline** (Tavily Search API), reducing zero-result queries by **60%** and ensuring hallucination-free medical data retrieval.
- Optimized relational **PostgreSQL** database schemas using **Prisma ORM** to efficiently manage 10,000+ records, improving query execution speeds by **35%** through strategic indexing and pagination.
- Engineered an **AI-driven OCR pipeline** for sub-2s medical prescription verification and successfully deployed fully responsive **Tailwind CSS** UI components via **Vercel CI/CD**.

OmniReach | SaaS Email Marketing (React, Node.js, AWS, Socket.io)

Mar. 2026 – Apr. 2026

- Created a scalable **email marketing SaaS** application using **React** and **Node.js**, incorporating **Google Gemini AI** to fully automate the generation of tailored HTML email templates.
- Orchestrated a real-time message delivery tracking system via **Socket.io** and **AWS SES** to ensure robust deliverability and provide users with instant campaign status analytics.
- Configured **AWS SNS** webhooks to handle the automated processing of bounce and complaint event metadata, alongside implementing a seamless 1-click unsubscribe mechanism.
- Implemented a background **node-cron** job scheduler with **MongoDB** persistence to manage asynchronous email dispatches, process **CSV** data streams, and compute 7-day analytics.

FoodieReels | Food Discovery Web App (React, Express.js, MongoDB)

Mar. 2026 – Apr. 2026

- Programmed a high-performance food discovery **REST API** utilizing **React**, **Express.js**, and **MongoDB**, featuring a fully customized **Gemini AI** chatbot for personalized user recommendations.
- Formulated an infinite-scroll feed utilizing Fisher-Yates deduplication, dual-tenant **RBAC** with **JWT**, and **ImageKit** **CDN** to ensure seamless media compression and highly optimized content delivery.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C/C++, Structured Query Language (SQL)

Frontend: React.js, Next.js, Redux, Tailwind CSS, shadcn/ui, HTML5, CSS3

Backend: Node.js, Express.js, RESTful APIs, Firebase, WebSockets, Socket.io, JWT

Databases: MongoDB, Mongoose ODM, PostgreSQL, Prisma ORM

AI & ML: Google Gemini AI, RAG Pipelines, OCR, Prompt Engineering, Tavily API

Tools & Cloud: Git/GitHub, AWS SES/SNS, Vercel, Docker, Postman, Figma, Android Studio, CI/CD Pipelines

Concepts & Core: System Design, Low-Level Design (LLD), Data Structures, Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS)

Achievements / Extracurricular

- Achieved **Global Rank 82** in **CodeChef Starters 240**; **4★** coder with highest rating **1822**.
- Guardian** on LeetCode with highest rating **2180**; secured **Rank 357** in Weekly Contest 505.
- Scored **10/10 SGPA** in 4th Semester at ABV-IIITM Gwalior.
- Completed **Graphs Camp** certification by **AlgoUniversity**.
- Merged **6 Pull Requests** across open-source repositories during **Hacktoberfest 2025**.